

Noel Sason

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EDUCATION

University of California, Berkeley — *B.A. Molecular Biology | B.S. Data Science*

Aug 2024 – May 2028

GPA: 3.82 / 4.00 | **Relevant Coursework:** CS61B (Data Structures & Algorithms), Data 100 (Principles & Techniques of Data Science), Math 54 (Linear Algebra), Data 8 (Foundations of Data Science — Statistics & Probability), Data C88C (Computational Structures in Data Science)

EXPERIENCE

Research Affiliate — *Lawrence Berkeley National Laboratory*

Jan 2026 – Present

- Built a DuckDB-backed data-modeling and analysis pipeline over the KG-Microbe knowledge graph, enabling fast SQL-based traversal across taxa, chemicals (CHEBI), environments (ENVO), anatomical sites (UBERON), and phenotypic traits for 80+ disease-microbiome studies.
- Wrote study-generation scripts that transform BugSigDB differential-abundance exports into per-study directories of increased/decreased taxa lists keyed to NCBI Taxon IDs, covering colorectal cancer, lung cancer, epilepsy, and skin wound cohorts.
- Designed a YAML-driven trait-profile generator that parses analysis results across all study directories and emits wide and long TSV tables (study × phenotype) for downstream statistical analysis and cross-cohort meta-analysis.
- Standardized reproducible workflows with Justfile task runners, uv-managed Python environments, and authored technical documentation; shipped via PR-based collaboration on GitHub with multiple merged contributions.

Finance Director — *Various Roles within the ASUC*

Sept 2024 – Present

- Finance Director for the South Asian Ad Hoc Committee and South Indian Society; administer approximately \$200,000 through CASSA, awarding grants to student-led proposals while ensuring ASUC financial compliance.

PROJECTS

Canvascope & Lectra — canvascope.org

Founder / CEO / Lead Developer

- Founded a privacy-first academic productivity platform spanning a Chrome extension (Manifest V3) and a native iOS companion app, scaling to 10+ university Canvas domains with local-first indexing and sub-100ms semantic search.
- Architected intent-aware semantic search over distributed course materials from a single query—fine-tuning embedding models in PyTorch and building the vector-search and ranking algorithm (cosine-similarity nearest-neighbor) with automated document categorization.
- Ran user interviews, usability studies, and A/B experiments with students across 10+ universities to study how people interact with the tool; analyzed and interpreted results to iterate on ranking, categorization, and UI.
- Developed Lectra, a native iOS/iPad app with Apple Pencil annotation and a custom DropBridge file-transfer protocol; built Supabase backend for real-time cross-device sync of annotated PDFs and course data.

RxBrief — rxbrief.org

Next.js, Javascript, Supabase

- Built a full-stack health platform featuring a live FDA drug label lookup API and real-time Supabase chat for patient-provider communication simulations.
- Designed client-server REST integrations for a multi-source healthcare jobs aggregator (Greenhouse, Lever, RSS APIs), implementing server-side filtering for rapid, domain-specific search results.
- Interviewed and collaborated with healthcare professionals to translate clinical workflows into product requirements and engineering tasks.

SIS Community App — *South Indian Society, UC Berkeley*

Swift, SwiftUI, Supabase

- Developing a native iOS app for a 200+ member student organization, featuring event management with Stripe payment integration, a shared photo album system backed by Cloudflare R2, and real-time messaging for member engagement.
- Architected a Supabase backend with role-based access control for leadership vs. general members, enabling committee-scoped file archives, push notifications, and centralized org operations replacing fragmented tools.

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Python, Java, Swift/SwiftUI, JavaScript, SQL, HTML5/CSS

ML & Data: PyTorch, scikit-learn, Pandas, NumPy, DuckDB, MongoDB, embeddings & vector search

Quant & Methods: Linear algebra, statistics & probability, A/B testing, data structures & algorithms

Web & Mobile: Node.js, Next.js, React, iOS/SwiftUI, Xcode, REST & client-server API design

Tools & Platforms: Git, GitHub Actions, Docker, Linux/Unix, Supabase, Vercel, Stripe, Cloudflare R2 **Cert:** AHA BLS